

Introduction

In the last couple of years, video creation has become an interesting topic of discussion among everyone, starting from creators to marketers. However, software engineers and designers have known about this convenient method of interaction for decades. It forms an essential part of their curriculum. As a student, it will be one of the essential components of your graphics syllabus too.

Today, we will talk about videos not from the perspective of using social media to garner reactions but from the point of view of an engineer or a designer. And hence, we shall get in touch upon a very important part of editing – both videos and images – called rendering.

What is rendering?

Rendering, also known as image synthesis, is used to create an image, either photorealistic or otherwise, from a premade 2D or 3D design, with the help of a code. The image produced as an outcome is known as the render. A scene file allows multiple files to be defined in strict languages like the data structure. It contains information related to viewpoint, geometry, lighting, texture and shading. This data is then sent to a program designed to render, which processes it and translates the output to a digital image. Another context where 'rendering' is used is in video editing. Rendering is the procedure of calculating effects in a video to generate the final footage. We will talk about this in the next section.

A picture that has been rendered will have the following features:

- **Shading:** Shading defines the change in brightness and colour of a surface with the shift in lighting.
- **Fogging:** Fogging defines the dimming of lights while passing through an unclear environment of air.
- **Bump-mapping:** Bump-mapping is the process of simulating bumps on various surfaces.
- **Texture-mapping:** Texture-mapping deals with the process of detailing the surfaces.
- **Shadows:** Like natural shadows, this effect deals with the obstruction of light.
- **Soft shadows:** Soft shadows are caused when restricted light sources cause darkness.

The other metrics are reflection, transparency or opacity, translucency,